

Project : Data Cleaning and Manipulation on goggle sheets

Task 1 : Identify the dirty elements in data and present data that is readable, accurate, and visually appealing.

- a. Rename your spreadsheet. Click **Untitled Spreadsheet**
- b. Make the columns wider by dragging the right boundary of the column heading. This may apply to the **name** (B), **address** (D), and **lat-long** (F) columns.
- c. Remove duplicates , there is at least one duplicate line (rows 20 and 21) in your dataset.
- d. Correct the rating data , ratings should fall between 0 and 5. However, at least one rating (in cell C8) falls outside of that range.
- e. Clean up longitude and latitude data ,data for latitude and longitude is contained in a single column (F), to use this data for analysis, the two values should be in separate columns.

Refer to the given link for : [Uncleaned Data](#)

Refer to the given link for : [Cleaned data my work](#)

Approaches i used are listed below :

Remove duplicates: The first step is to eliminate any duplicate entries from your dataset. As a best practice, duplicates should be removed even if they are not readily apparent.

1. To start, select columns A through F.
2. Then,choose **Data**, then **Data Cleanup**, and select **Remove duplicates**.
3. click **Data has header row**. You want to remove duplicate boba shop id's and boba shop names, made sure the relevant columns (**id**, **name**) are selected.

Correct the rating data:

- a. Yelp ratings should be less than 5 and greater than 0. determined how many entries are inaccurate and correct them. used the **COUNTIF** function to perform this task.

1. In cell I2, enter `=COUNTIF(C:C, ">5")`. The first entry (**C:C**) the range is the entire **rating** column (C), which contains the Yelp ratings. The second entry refers to the criterion (**>5**), and tells the function to count all the values greater than 5.
2. Press **Enter**. the function returns a value of 9. dataset contains 9 entries that have a rating greater than 5.
3. Select columns A through F.
4. Then, from the menu bar, choose **Data**, then **Sort range**, and select **Advanced range sorting options**.
5. In the pop-up window, check the box next to **Data has header row**. Sort by **rating** from **Z → A**. This way, the highest ratings will be listed first.
6. Click **Sort**. Check out your spreadsheet. At the start of the **rating** column, you should now find the 9 rows that have incorrect values (rating > 5).
7. Next, select the range of cells **C2:C10**. Press **delete** to delete the values that are greater than 5.
8. Replace all the values with the number **5**. In cell C2, enter **5**. Then, drag the fill handle down to cell C10 to fill the remaining cells with **5**.
9. After replacing the incorrect ratings with the number 5, you may notice that the new value in cell I2 is 0. The output of the `COUNTIF` function now reflects the changes in your dataset. This confirms that the **rating** column no longer contains any values greater than 5.
10. delete the formula from cell I2 since you don't need this information anymore.

Cleanup the longitude and latitude data:

1. The **SPLIT** function divides text around a specified character or string, and puts each fragment of text into a separate cell in the row. The **SPLIT** function will split the single **lat-long** column into two separate columns, one for latitude and the other for longitude. In cell G2, enter **=SPLIT(F2, "-")**. The first entry (**F2**) refers to the cell where the text is located. The second entry ("**-**") refers to the fact that you are dividing the text based on the minus sign.
2. Press **Enter**. The result shows each fragment of text in a different cell.
3. Select cell G2 again. In cell G2, double-click on the fill handle to split all the remaining **lat-long** entries.
4. Now add column headers to the two new columns (G and H). In cell G1, enter **lat**. In cell H1, enter **long**.
5. Next, replace the original **lat-long** data in column F with the new split entries in columns G and H. Select columns G and H, right-click, and choose **Copy**.
6. Then, select Column F, right-click, and choose **Paste special** and **Paste values only**.
7. Now the new **lat** column is column F, and the new **long** column is column G. Adjust the width of the **lat** column (F) to fit the data by dragging the right boundary of the column heading.
8. Next, select column H, right-click, and choose **Delete column**.
9. Finally, the longitude values should be negative so that they are accurate coordinates for mapping. To make the values in the **long** column negative, multiply them by **-1**. In cell H2, enter **=G2*-1**. The asterisk is the operator for multiplication. Press **Enter**.
10. Still in cell H2, double-click on the fill handle to fill in the rest of the values.
11. Next, add a column header. In cell H1, enter: **long**.

12. Now, replace the longitude data in column G with the new data in column H. Select column H, right-click, and choose **Copy**.

13. Select Column G, right-click, and choose **Paste special** and **Paste values only**.

14. Then, select column H, right-click, and choose **Delete column**.